Post-Submission Questionnaire & Answers

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1. Thought Process

Problem breakdown

I separated the workflow into modules: input parsing and validation, engineering calculations, CAD generation, drawing export, BOM and costing, and manufacturing timeline. A single orchestrator then runs all modules in sequence.

Automated parts and why

I automated requirement parsing, engineering calculations, CAD generation, drawing generation, and BOM/costing because these are repetitive and rule-based. Automation reduces manual errors and ensures repeatability.

Trade-offs between accuracy and speed

I used deterministic first-pass formulas and conservative safety factors instead of full FEA. This gives fast turnaround and clear assumptions, suitable for concept-level design in a 72-hour timeline.

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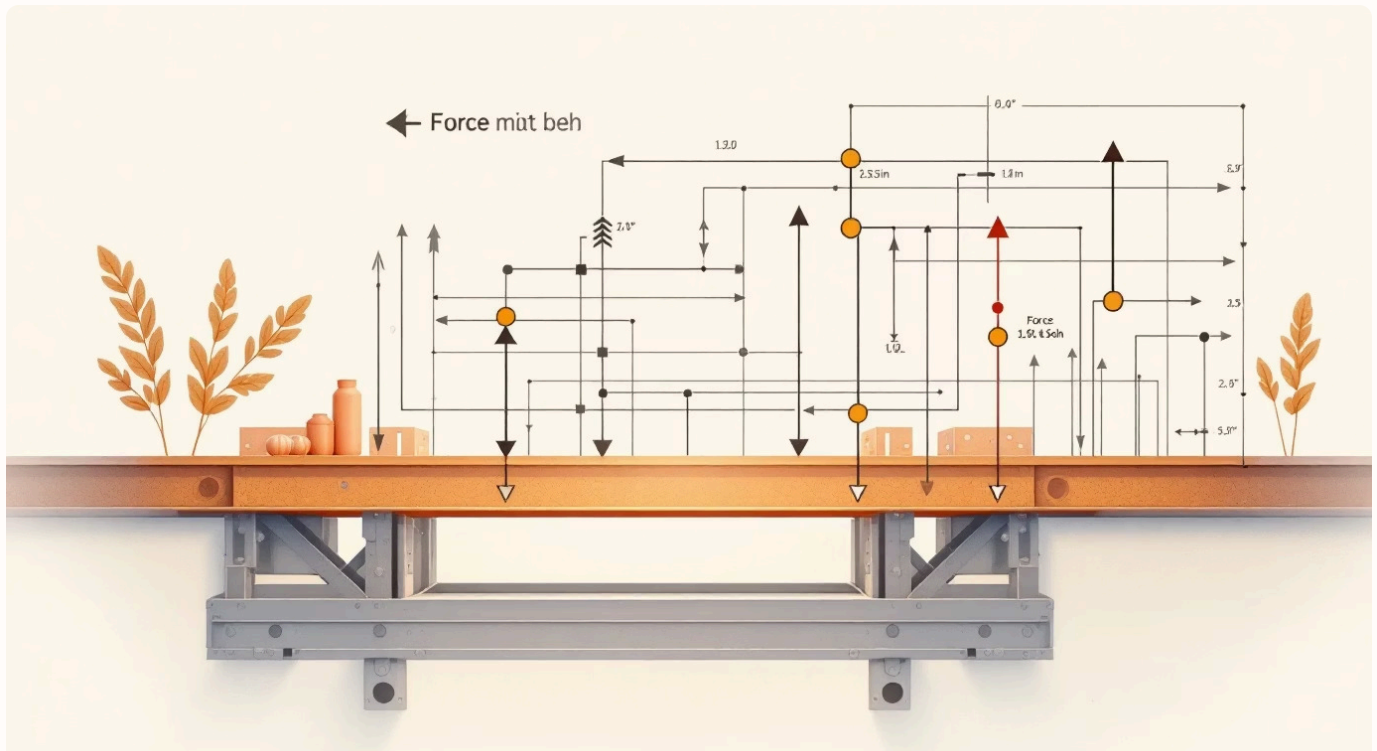
2. Engineering Decisions

Assumptions

- Material strength: Yield strengths assumed from standard reference values.
- Load distribution: Uniform/centralized distribution for first-pass calculations.
- Safety factor: 2.5 for industrial handling context.

Weight and dimensions method

Frame mass is estimated from selected tube section and total frame member length. Deck mass is computed from plate volume and material density. Key dimensions are input-driven with rule-based member sizing.



Improvements with more time

I would add dynamic load factors, finite element checks, wheel bearing life checks, and richer CAD assembly constraints.

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3. Automation Approach

Where automation is used

- Requirement parsing
- Engineering calculations
- CAD script generation
- Drawing export
- BOM/costing generation

AI tool usage

AI was used for workflow decomposition, documentation drafting support, and sanity-checking calculation assumptions. Final logic and formulas were implemented and validated in code.

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4 . CAD and Design

01

CAD tool choice

I chose an OpenSCAD-script path for reliable, free, script-first generation under time constraints, while keeping the architecture compatible with FreeCAD migration.

02

Parameterization

Model geometry is parameterized by input dimensions, wheel count, environment-linked wheel size, and computed platform thickness.

03

CAD automation limitations

Current script creates concept-level geometry and does not include detailed weld features or tolerance stack-up.

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5. Costing and Manufacturing

Material cost estimation

Material cost is estimated from computed mass multiplied by INR/kg assumptions, then combined with wheel and fastener assumptions.

Fabrication time assumptions

Fabrication and assembly durations are baseline hours scaled by load and platform size factors.

Cost realism

The estimate is reasonable for early-stage quotation but requires supplier quotes and process routing for production-level accuracy.

Timeline clarification

The project execution window is 72 hours as per assessment instructions. The manufacturing timeline in outputs is a separate estimate for fabrication and assembly effort of the trolley, not the assignment duration.

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6. Challenges and Debugging

Biggest challenges

Balancing implementation speed with engineering clarity and ensuring every mandatory output is auto-generated in one flow.

Two specific problems and fixes

- Problem: Need for free and scriptable CAD output under time constraint. Fix: OpenSCAD script generator as robust baseline.
- Problem: PDF drawing dependency uncertainty. Fix: Optional matplotlib path with fallback text output.

Most fragile part

Rendering-level drawing export is the most environment-sensitive component because it depends on plotting libraries.

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7. Improvements

If given 2 more weeks

I would implement the following six upgrades:

1. Integrate an LLM-based requirement parser to convert free-text client requests (also the type of design of the trolley the client wants) into validated structured inputs.
2. Integrate a cloud CAD API (for example Onshape API) to generate more production-ready parametric geometry and drawing views.
3. Add simulation API integration for stress/deflection sanity checks before finalizing the design output.
4. Integrate live material and supplier pricing APIs to replace static INR per kg assumptions with time-stamped market data.
5. Add supplier catalog API integration for wheel and fastener selection based on load ratings and availability.
6. Integrate ERP/inventory API mapping so BOM items can be checked against stock and lead-time constraints.

Scaling to industrial use

Add a material database, supplier API integration, revision control, simulation gates, and ERP-ready BOM export.

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8. Self-Assessment

What was done well

Clear modular workflow, reproducible outputs, and direct traceability from input to calculations and deliverables.

Weak points

Concept-level structural assumptions and simplified CAD detail; simulation and standards compliance checks are future work.

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Summary

Modular Workflow

Clear separation of concerns with input parsing, calculations, CAD generation, and costing modules orchestrated sequentially.

Automation Focus

Rule-based processes automated to reduce errors and ensure repeatability across requirement parsing, calculations, and documentation.

Concept-Level Design

Deterministic formulas with conservative safety factors provide fast turnaround suitable for 72-hour timeline constraints.

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